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# Effects of benfotiamine treatment over 12 months on morphometric, neurophysiological and clinical measures in type 2 diabetes patients with symptomatic polyneuropathy: a randomized, placebo-controlled, double-blind clinical trial (BOND study)

Dan Ziegler ,[1] Gundega Sipola ,[1] Alexander Strom,[1,2] Klaus Strassburger ,[2,3] Luise Jander,[3] Christian Herder ,[1,2,4] Birgit Knebel ,[2,5] Claudia Reule ,[6] Benjamin Assad Jaghutriz,[6] Andrea Carolina Moreira Tupac Yupanqui ,[6] Andrea Icks ,[2,7,8] Hadi Al-Hasani ,[2,5] Michael Roden ,[1,2,4] Oliver Kuss ,[2,3,9] Gidon J Bönhof [1,2,4]

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For numbered affiliations see end of article.

**Correspondence to**
Professor Dan Ziegler;
dan.ziegler@ddz.de

## ABSTRACT

**Introduction** Shunting of glycolytic intermediates into the pentose phosphate pathway via transketolase activation by benfotiamine has been suggested to protect from hyperglycemia-induced microvascular damage, but the long-term effects of benfotiamine on diabetic sensorimotor polyneuropathy (DSPN) remain unclear.

**Research design and methods** This 1:1 randomized double-blind, placebo-controlled parallel group monocentric phase II trial compared the efficacy and safety of 1-year treatment with benfotiamine 300 mg two times per day versus placebo over 12 months in participants with type 2 diabetes and mild-to-moderate symptomatic DSPN. The primary endpoint was the change in corneal nerve fiber length (CNFL) assessed by corneal confocal microscopy (CCM) from baseline to 12 months. Secondary endpoints included three other CCM parameters, skin biopsy (four parameters), nerve conduction studies (13 measures), quantitative sensory testing (six parameters), cardiovascular autonomic function tests (17 indices), sudomotor function tests (five parameters), 15 clinical scores and scales for neuropathic symptoms and signs and 13 health-related quality of life and depression instruments. Pharmacokinetics included measurement of six thiamine analytes in blood.

**Results** A total of 57 participants underwent randomization. The changes from baseline to 12 months in CNFL did not differ between the two groups. The corresponding changes in the secondary morphometric, functional and clinical neuropathic outcomes as well as quality of life were also similar in the two groups. Only the Neuropathy Symptom Score tended to improve after benfotiamine treatment (p=0.098 vs placebo). Benfotiamine treatment increased the concentrations of all six thiamine analytes studied (p≤0.003 vs placebo). Safety

### WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Shunting of glycolytic intermediates into the pentose phosphate pathway via transketolase activation by benfotiamine has been suggested to protect from hyperglycemia-induced microvascular damage. However, the long-term effects of benfotiamine, a lipid-soluble prodrug of thiamine (vitamin B1) and transketolase activator, on diabetic sensorimotor polyneuropathy remain unclear.

### WHAT THIS STUDY ADDS

⇒ In type 2 diabetes individuals with mild-to-moderate symptomatic polyneuropathy, treatment with benfotiamine for 12 months was well tolerated, but had no significant effects on multiple morphometric, neurophysiological and clinical measures of neuropathy.

### HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ These findings do not contradict the use of benfotiamine for treatment of neuropathies caused by vitamin B1 deficiency, but favorable treatment effects within 12 months on objective neuropathic outcomes in individuals with type 2 diabetes, unselected for vitamin B1 levels, seem unlikely.

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analysis showed no relevant differences between the groups in the rates of adverse events.

**Conclusions** In type 2 diabetes individuals with mild-to-moderate symptomatic DSPN, treatment with benfotiamine for 12 months was well tolerated, but had no significant effects on multiple morphometric, neurophysiological and clinical measures of neuropathy.

**Trial registration number** European Clinical Trials Database (EudraCT) 2017-003054-16 registered on April 10 (https://eudract.ema.europa.eu/), 2018 and German Register for Clinical Trials DRKS00014832 registered on August 3, 2018 (https://drks.de/search/de).

## INTRODUCTION

Diabetic sensorimotor polyneuropathy (DSPN) affects approximately one-third of community-based samples of people with diabetes. It is associated with considerable morbidity and markedly impaired quality of life by causing neuropathic pain and foot ulcers and carries an increased risk of mortality.[1 2] Thus, DSPN is responsible for a high burden on the healthcare system and society[3] but remains underdiagnosed and undertreated.[4 5] Several analgesic agents for the pharmacotherapy of painful DSPN are available, but these drugs only alleviate neuropathic pain and do not favorably influence the underlying neuropathy. Therefore, pathogenesis-derived pharmacotherapies have been developed aimed not only at improving symptoms but also at slowing or even reversing the development or progression of DSPN.[2]

Benfotiamine, a lipid-soluble prodrug of thiamine (vitamin B1) and transketolase activator with a high bioavailability, was shown in diabetic rodents to block the flux through three major pathways (the hexosamine pathway, the advanced glycation end product (AGE) formation pathway, and the diacylglycerol-protein kinase C pathway), which are activated by increased availability of the glycolytic metabolites glyceraldehyde-3-phosphate and fructose-6-phosphate and are implicated in the pathogenesis of diabetic microvascular complications.[6 7] Moreover, benfotiamine exerts direct antioxidant properties.[8 9] In experimental diabetic neuropathy, benfotiamine nearly normalized nerve conduction velocity (NCV), induced an inhibition of neural imidazole-type AGE formation and completely prevented diabetes-induced glycoxidation products after 6 months,[10] thus providing a rationale for the treatment of DSPN in humans.[11]

In people with diabetes and healthy smokers, benfotiamine was shown to improve endothelial function.[12 13] Several studies conducted over periods up to 12 weeks indicated that benfotiamine alone or in combination with vitamin B6 and B12 may alleviate neuropathic symptoms in patients with DSPN.[14–17] Except for one short-term trial over 12 weeks demonstrating that benfotiamine in combination with vitamin B6 and B12 improved NCV,[14] no study in people with diabetes is available showing that benfotiamine may favorably modify the natural history of DSPN.

Thus, there is an unmet need for longer-term studies to demonstrate that benfotiamine offers a pathogenesis-derived approach that may delay or halt the clinical, functional or morphological progression of DSPN. The present **B**enfotiamine **O**n morphometric, **N**europhysiological and clinical measures in type 2 **D**iabetes patients (BOND) trial is a randomized double-blind, placebo-controlled parallel group monocentric phase II clinical trial to assess the benefits and harms of treatment with benfotiamine compared with placebo in participants with type 2 diabetes and mild to moderate symptomatic DSPN.[18] Since it remains unclear which components of DSPN may be most susceptible to benfotiamine treatment, the BOND study includes an array of clinical, functional and morphological outcomes with emphasis on small fiber neuropathy as well as several safety parameters.

## METHODS

### Study design

The study design and methodology have been published in detail elsewhere.[18] In brief, the BOND study was a randomized, double-blind, placebo-controlled, parallel group proof-of-concept, exploratory phase II trial to assess the effects of treatment with benfotiamine for 12 months on morphometric, neurophysiological, clinical and health-economic measures in participants with type 2 diabetes and mild to moderate symptomatic DSPN. The latter was defined by the presence of neuropathic symptoms for ≥6 months and a Neuropathy Disability Score (NDS) of 3–8 points confirmed by at least one of the following: reduced sural sensory NCV, sural sensory nerve action potential, peroneal motor NCV (MNCV) and/or tibial MNCV. The trial was conducted as a mono-center trial at the Institute for Clinical Diabetology, German Diabetes Center (DDZ) at Heinrich Heine University (HHU) in Düsseldorf, Germany, in accordance with the principles of the Declaration of Helsinki. The trial received a favorable opinion by the Ethics Committee at the HHU Faculty of Medicine, Düsseldorf, Germany, on 27 July 2018 and was approved by the German Federal Institute for Drugs and Medical Devices on 8 June 2018. Patients were not involved in the design, conduct or reporting of this study. However, the trial included a broad set of patient-reported outcomes to capture the patient perspective. During the trial, inclusion and exclusion criteria were adapted to improve recruitment and participant safety. All changes were approved by the ethics committee and regulatory authority.

### Participants and interventions

After screening for eligibility, participants were randomly assigned 1:1 using a computer-generated randomization sequence to oral treatment with benfotiamine 300 mg (Milgamma mono 300; Wörwag Pharma, Böblingen, Germany) or matching placebo, two times per day for 12 months. The randomization list was provided exclusively to the packaging site, with a backup retained by the independent biometrician, not involved in efficacy analysis.

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Participants, care providers, outcome assessors, data analysts and sponsors remained completely blinded until database lock. Participants were assigned a participant number in consecutive order and received the treatment that has been allocated to that number as given by the randomization code. Trial staff involved in allocation had no access to the randomization code. Participants were recruited between 18 October 2018 (first patient first visit) and 22 February 2023 (last patient first visit), with final follow-up completed on 13 March 2024 (last patient last visit). A flow diagram of the participants' progress through the trial is illustrated in online supplemental figure 1.

### Outcomes

The primary focus was to study the potential of benfotiamine to induce nerve regeneration, which can be quantified by skin biopsy and CCM. At the time at the time of completing the BOND trial design, no randomized controlled trials (RCTs) using these methods were available to calculate the sample size needed. Therefore, a prospective observational study using CCM over 1 year served to this end.[18 19] Accordingly, the primary endpoint was the difference in the change in CNFL assessed by CCM between the groups treated with benfotiamine and placebo after 12 months. Secondary endpoints included three other CCM parameters, skin biopsy (four parameters), nerve conduction studies (NCS; 13 parameters), quantitative sensory testing (QST; six parameters), cardiovascular autonomic function tests (CAFTs; nine parameters), spontaneous baroreflex sensitivity (BRS; eight parameters), sudomotor function tests (five parameters), 15 clinical scores and scales for neuropathic symptoms and signs and three health-related quality of life and depression instruments.

Vitamin B1 analytes (thiamine diphosphate (TDP), thiamine monophosphate (TMP), free thiamine, total thiamine, transketolase activity) were determined using high-pressure liquid chromatography with fluorescence detection (HPLC-FD) with an Agilent HPLC-System 1260. The thiamine pyrophosphate (TPP) effect was measured by photometry using a Hitachi Fotometer U 1800/U 2900 (Laboratory Lademannbogen, Hamburg, Germany). Adverse events, vital signs and safety laboratory parameters were analyzed to assess potential harms systematically. Treatment compliance was monitored through tablet counts.

### Statistical analysis

Continuous variables were summarized with descriptive statistics by displaying sample size, mean, median, minimum, maximum, SD and 95% CI. Categorical data were summarized using frequencies and percentages. Differences of baseline to post-treatment changes after 12 months between treatment arms were analyzed by means of analysis of covariance (ANCOVA), where the post-treatment outcome enters the model as the dependent variable and a treatment indicator and the baseline outcome, age, sex, COVID-19 information (pre, during and post-COVID-19 pandemic) and concomitant treatment with gabapentin, pregabalin or duloxetine enter as independent variables. The level of statistical significance was set at $\alpha$=0.05. Sample size calculation was based on CCM as the primary endpoint, based on a prospective observational study[19] assuming an increase in CNFL of 4.85±5.71 (mean±SD) mm/mm$^2$.[18] Harms were assessed in all participants receiving at least one dose of treatment, systematically using CTCAE criteria and analyzed descriptively across treatment groups. Frequencies and percentages of TEAEs and several serious adverse events (SAEs) were tabulated, including severity and causality ratings. No formal statistical testing was applied to harms.

The primary analysis was performed in the intention to treat (ITT) population. Missing values were imputed by multiple imputation technique. The ITT population was defined as all participants having at least one dose of treatment and at least one measurement of the primary endpoint under treatment. The safety analysis set included all participants who received at least one dose of treatment. Parameters for safety analysis included adverse events, concomitant medication, laboratory safety variables, physical examination and vital signs. Efficacy was analyzed both in ITT and PP populations (not reported). Within the ITT set, sensitivity analyses using ANCOVA with multiple imputation addressed missing data. Prespecified subgroup analyses were planned but not yet performed.

## RESULTS

### Efficacy analysis

The demographic and clinical characteristics at baseline of the full analysis/safety data set are shown in online supplemental Table 1. The levels of HbA1c, weight, waist circumference, blood pressure and heart rate during the study are shown in online supplemental Table 2. There were no significant differences between the groups in the changes of these variables from baseline to 6 and 12 months.

Table 1 depicts the results for the morphometric measures derived from CCM and skin biopsy at baseline and after 6 and 12 months with the corresponding differences from baseline to 12 months and treatment effects. Neither the changes from baseline to 12 months in corneal nerve fiber length (CNFL) as the primary endpoint, nor those in the remaining three CCM measures differed significantly between the groups. The observed increase in CNFL from baseline to 12 months in participants treated with benfotiamine was minimal and, thus, far below the anticipated effect size. Likewise, neither the changes from baseline to 12 months in intraepidermal nerve fiber density (IENFD) nor those in the measures obtained by CD31, SOD2 and CD207 antibody staining showed significant differences between the groups.



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**Table 1** Morphometric measures derived from corneal confocal microscopy (CCM) and skin biopsy

| Endpoint | Treatment groups (n) | Baseline (BL) | Month 6 | Month 12 | Difference BL vs month 12 | Treatment effect (95% CI) | P value |
|---|---|---|---|---|---|---|---|
| ***CCM*** | | | | | | | |
| CNFL (mm/mm²) | Benfotiamine (24)<br>Placebo (27) | 15.650 (4.410)<br>16.868 (3.794) | 17.263 (5.494)<br>18.615 (6.103) | 16.192 (4.905)<br>17.875 (6.158) | 0.559 (3.581)<br>0.835 (3.544) | −0.327 (−2.432 to 1.778) | 0.760 |
| CNFD (mm/mm²) | Benfotiamine (24)<br>Placebo (27) | 18.749 (4.397)<br>17.573 (4.186) | 18.116 (4.337)<br>19.406 (5.451) | 17.858 (5.397)<br>18.315 (6.034) | −0.938 (4.206)<br>0.497 (4.616) | −0.935 (−3.529 to 1.658) | 0.479 |
| CNBD (number/mm²) | Benfotiamine (24)<br>Placebo (27) | 33.853 (16.852)<br>34.624 (15.890) | 42.481 (22.021)<br>42.978 (30.575) | 36.931 (20.064)<br>44.574 (25.889) | 3.220 (14.991)<br>10.396 (19.709) | −5.370 (−15.73 to 4.990) | 0.309 |
| CNFT | Benfotiamine (24)<br>Placebo (27) | 0.173 (0.045)<br>0.173 (0.062) | 0.179 (0.036)<br>0.179 (0.025) | 0.170 (0.038)<br>0.165 (0.039) | −0.001 (0.056)<br>−0.010 (0.069) | 0.007 (−0.015 to 0.030) | 0.531 |
| ***Skin biopsy*** | | | | | | | |
| IENFD PGP9.5 (number/mm) | Benfotiamine (24)<br>Placebo (27) | 3.469 (3.615)<br>3.145 (3.013) | n.a. | 1.730 (2.407)<br>2.750 (3.057) | −1.435 (2.040)<br>−0.792 (2.969) | −0.486 (−1.870 to 0.897) | 0.484 |
| CD31 (%) | Benfotiamine (24)<br>Placebo (27) | 2.329 (0.946)<br>2.164 (0.777) | n.a. | 2.214 (0.982)<br>2.366 (1.064) | −0.253 (1.342)<br>0.194 (1.343) | −0.110 (−0.883 to 0.663) | 0.775 |
| SOD2 (%) | Benfotiamine (24)<br>Placebo (27) | 0.514 (0.198)<br>0.523 (0.170) | n.a. | 0.388 (0.215)<br>0.421 (0.222) | −0.126 (0.127)<br>−0.089 (0.184) | −0.016 (−0.137 to 0.106) | 0.792 |
| CD207 (number/mm²) | Benfotiamine (24)<br>Placebo (27) | 194.864 (85.493)<br>183.115 (52.770) | n.a. | 157.350 (98.445)<br>166.889 (65.668) | −49.100 (118.729)<br>−20.941 (52.380) | −12.87 (−65.70 to 39.957) | 0.630 |

CD31, endothelial cell area-cluster of differentiation 31; CD207, Langerhans cell density-cluster of differentiation 207; CNBD, corneal nerve branch density; CNFD, corneal nerve fiber density; CNFL, corneal nerve fiber length; CNFT, corneal nerve fiber tortuosity; IENFD, intraepidermal nerve fiber density; n.a., not assessed; SOD2, mitochondrial superoxide dismutase area.



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No significant differences between the groups were observed in the changes from baseline to 12 months for any of the 13 NCS measures (table 2) and for the six QST parameters (table 3).

Table 4 summarizes the outcomes for the various clinical scores for neuropathic symptoms and signs. Overall, no significant differences between the groups were observed for the changes in each of the scores from baseline to 12 months. Only the Neuropathy Symptom Score (NSS) was reduced by trend after 12 months in the group treated with benfotiamine as compared with the placebo group (p=0.098).

Table 5 shows the blood concentrations of thiamine analytes over 12 months. TDP, TMP and free and total thiamine increased (all p≤0.003), while the TPP effect decreased (p=0.002) from baseline to 12 months in the group treated with benfotiamine as compared with the group administered placebo (TDP: +206% vs +22%, TMP: +220% vs +11%, free thiamine: +3550% vs +302%, total thiamine: +94% vs +21%, transketolase: +21% vs −5%, TPP effect: −77% vs −2.6%).

No significant differences were detected in the changes from baseline to 12 months between the groups for CAFTs (nine indices) and BRS (eight indices) (online supplemental Table 3), sudomotor function studies assessed by the sweat indicator test Neuropad and electrochemical skin conductance quantified by the Sudoscan device (online supplemental Table 4). The qualitative Neuropad test results yielded an OR of 1.103 (95% CI 0.324 to 3.759) for benfotiamine versus placebo after 12 months (p=0.875).

None of the scales or subscales of the EuroQol 5-Dimension 5-level questionnaire, 36-Item Short Form Survey and Patient Health Questionnaire-9 showed significant differences in the changes from baseline to 12 months between the groups (online supplemental Table 5).

### Safety analysis

Within the study, 27 participants (96.4%) in the benfotiamine group and 27 participants (93.1%) in the placebo group experienced treatment emergent adverse events (TEAEs). The benfotiamine group had a total of 229 TEAEs, while the placebo group had 140 TEAEs. Five participants in the benfotiamine group had considerably more TEAEs, including SAEs, which were rated as not related or unlikely related to the study drug. The benfotiamine group had 19 SAEs (8.3% of TEAEs) in six participants (21.4%), with most SAEs being of moderate intensity (89.5%). The placebo group had 9 SAEs (6.4% of TEAEs) in five participants (17.2%), with SAEs classified as mild (33.3%), moderate (33.3%) and severe (33.3%). None of the SAEs was rated as related to study drug by the investigator. The rates of TEAEs for each primary system organ class (primSOC) are listed in online supplemental Table 6. TEAEs rated as probably or possibly related to study drug were more common in the placebo group (placebo: 6.9%, four participants vs benfotiamine: 3.6%, two participants). The compliance of the investigational medicinal product intake was >80% for all completing participants, and safety laboratory parameters did not show major differences between treatment groups. Concomitant medications were commonly used in both groups, primarily targeting metabolic, cardiovascular and hematologic conditions, with similar patterns observed across the treatment arms.

## DISCUSSION

The ITT analysis of this phase II monocenter, randomized, double-blinded, placebo-controlled trial showed that treatment with benfotiamine administered 300 mg orally two times per day over 12 months, independent of any vitamin B1 deficiency, had no effect on the primary morphometric outcome (CNFL) in type 2 diabetes individuals unselected for vitamin B1 levels with mild-to-moderate symptomatic DSPN. Furthermore, there were no significant effects on the secondary morphometric, functional and clinical neuropathic endpoints as well as quality of life when compared with placebo treatment. One trend of borderline statistical significance in favor of benfotiamine was found for the NSS in the lower limbs. Participants treated with benfotiamine showed a distinct increase in thiamine analytes in blood after 12 months, representing a markedly augmented thiamine supply. Safety analysis confirmed that the drug was well-tolerated.

The lack of effect of benfotiamine on a wide array of DSPN outcomes after 1 year needs to be discussed in light of the findings of previous trials in patients with DSPN. Altogether, four randomized, double-blind controlled clinical trials conducted over 3–52 weeks assessed the efficacy and safety of different daily doses of benfotiamine in patients with DSPN primarily focusing on neuropathic symptoms rather than objective measures of nerve function and morphology.[14 15 17 20] In the 6-week Benfotiamine in Diabetic Polyneuropathy (BENDIP) study, 165 patients with DSPN received either benfotiamine 300 mg two times per day, 100 mg three times per day or placebo. After 6 weeks, the NSS as the primary endpoint did not improve with benfotiamine versus placebo in the ITT analysis, whereas the PP analysis showed significant improvement for the 300 mg two times per day (p=0.033) but not for the 100 mg three times per day dose. The secondary endpoints (Total Symptom Score; TSS, NDS, vibration perception threshold; VPT) did not differ between the groups.[17] Likewise, a borderline positive effect on the NSS but lacking effect on the TSS was observed in the present study after 12 months. In the 3-week Benfotiamine in the Treatment of Diabetic Polyneuropathy (BEDIP) study, 40 patients with DSPN were administered benfotiamine 100 mg four times per day or placebo. After 3 weeks, the primary endpoint, a composite score of muscle strength, pain, sensory function, coordination and reflexes was significantly improved with benfotiamine as compared with placebo.[15]



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**Table 2** Nerve conduction studies

| Endpoint | Treatment groups (n) | Baseline (BL) | Month 6 | Month 12 | Difference BL vs month 12 | Treatment effect (95% CI) | P value |
|---|---|---|---|---|---|---|---|
| Median MNCV (m/s) | Benfotiamine (24)<br>Placebo (27) | 52.783 (8.496)<br>49.754 (7.921) | 52.996 (5.297)<br>49.573 (9.762) | 51.900 (4.490)<br>49.510 (10.248) | −0.160 (9.123)<br>−1.814 (7.543) | 1.048 (−3.334 to 5.430) | 0.639 |
| Median SNCV (m/s) | Benfotiamine (24)<br>Placebo (27) | 47.831 (13.191)<br>48.246 (9.368) | 54.267 (4.015)<br>40.700 (18.750) | 48.680 (17.500)<br>44.567 (12.791) | −2.214 (21.752)<br>1.400 (3.006) | −1.381 (−10.75 to 7.991) | 0.771 |
| Median SNAP (µV) | Benfotiamine (24)<br>Placebo (27) | 6.133 (5.082)<br>5.662 (3.328) | 6.492 (5.180)<br>4.345 (2.947) | 6.420 (3.278)<br>5.511 (3.576) | 0.133 (3.347)<br>−0.540 (2.631) | 0.466 (−1.956 to 2.888) | 0.685 |
| Ulnar MNCV (m/s) | Benfotiamine (24)<br>Placebo (27) | 52.548 (12.155)<br>53.250 (8.764) | 53.138 (6.796)<br>51.669 (10.010) | 53.176 (7.565)<br>53.490 (5.587) | 2.255 (9.581)<br>−0.786 (5.954) | 0.008 (−3.932 to 3.948) | 0.997 |
| Ulnar SNCV (m/s) | Benfotiamine (24)<br>Placebo (27) | 45.994 (16.872)<br>50.569 (16.769) | 55.678 (7.307)<br>44.892 (19.143) | 50.357 (23.082)<br>52.400 (10.521) | 4.133 (26.293)<br>−7.183 (6.980) | −2.119 (−13.63 to 9.389) | 0.714 |
| Ulnar SNAP (µV) | Benfotiamine (24)<br>Placebo (27) | 7.994 (13.013)<br>4.823 (2.640) | 4.644 (2.538)<br>4.092 (2.769) | 5.514 (3.242)<br>4.914 (3.408) | −9.550 (19.590)<br>−1.450 (2.041) | −0.430 (−3.266 to 2.407) | 0.755 |
| Peroneal MNCV (m/s) | Benfotiamine (24)<br>Placebo (27) | 39.675 (5.107)<br>38.370 (4.019) | 37.846 (6.449)<br>36.819 (5.432) | 35.929 (7.083)<br>38.114 (7.281) | −3.243 (5.132)<br>−0.273 (5.737) | −2.892 (−6.527 to 0.744) | 0.118 |
| Tibial MNCV (m/s) | Benfotiamine (24)<br>Placebo (27) | 33.900 (8.156)<br>32.919 (5.458) | 33.743 (6.996)<br>31.012 (4.554) | 31.432 (6.042)<br>31.370 (6.247) | −1.222 (7.845)<br>−0.840 (7.144) | −0.147 (−3.815 to 3.521) | 0.937 |
| Sural SNCV (m/s) | Benfotiamine (24)<br>Placebo (27) | 28.757 (7.694)<br>27.763 (5.272) | 29.613 (9.268)<br>26.944 (4.891) | 25.948/ (5.132)<br>26.048 (3.369) | −1.805 (9.046)<br>−1.376 (3.889) | −0.618 (−3.198 to 1.962) | 0.639 |
| Sural SNAP (µV) | Benfotiamine (24)<br>Placebo (27) | 1.735 (3.002)<br>2.341 (4.760) | 1.771 (2.181)<br>1.211 (2.186) | 1.533 (2.975)<br>1.305 (2.568) | −0.085 (3.051)<br>−1.224 (5.551) | −0.414 (−1.779 to 0.950) | 0.551 |
| MNCV sum score | Benfotiamine (24)<br>Placebo (27) | 0.101 (0.815)<br>−0.092 (0.617) | 0.151 (0.604)<br>−0.145 (0.607) | −0.005 (0.625)<br>0.003 (0.758) | 0.030 (0.608)<br>0.058 (0.626) | −0.037 (−0.420 to 0.347) | 0.850 |
| SNCV sum score | Benfotiamine (24)<br>Placebo (27) | −0.103 (0.873)<br>−0.041 (0.670) | 0.093 (0.722)<br>−0.296 (0.486) | −0.066 (0.789)<br>−0.044 (0.548) | 0.180 (1.118)<br>−0.031 (0.576) | −0.022 (−0.444 to 0.400) | 0.919 |
| NCV sum score | Benfotiamine (24)<br>Placebo (27) | 0.020 (0.711)<br>−0.070 (0.569) | 0.161 (0.536)<br>−0.176 (0.509) | −0.030 (0.597)<br>−0.001 (0.622) | 0.081 (0.620)<br>0.041 (0.501) | 0.005 (−0.345 to 0.355) | 0.979 |

MNCV, motor nerve conduction velocity; NCV, nerve conduction velocity; SNAP, sensory nerve action potential; SNCV, sensory nerve conduction velocity.



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Table 3 Quantitative sensory testing

| Endpoint | Treatment groups (n) | Baseline (BL) | Month 6 | Month 12 | Difference BL vs month 12 | Treatment effect (95% CI) | P value |
|---|---|---|---|---|---|---|---|
| Metacarpal VPT (µm) | Benfotiamine (24) | 1.464 (1.144) | 1.183 (0.902) | 1.843 (1.933) | 0.280 (2.000) | 0.323 (-0.684 to 1.330) | 0.526 |
| | Placebo (27) | 1.858 (2.811) | 1.637 (1.440) | 1.802 (2.993) | −0.092 (1.004) | | |
| Malleolar VPT (µm) | Benfotiamine (24) | 9.202 (5.471) | 8.397 (5.689) | 7.398 (4.912) | −1.948 (4.303) | −0.639 (-3.099 to 1.822) | 0.609 |
| | Placebo (27) | 8.392 (5.809) | 8.237 (6.058) | 7.797 (5.318) | −1.058 (4.976) | | |
| CDT thenar (°C) | Benfotiamine (24) | 29.400 (2.110) | 28.970 (2.119) | 29.310 (1.339) | −0.021 (1.484) | −0.227 (-0.806 to 0.352) | 0.437 |
| | Placebo (27) | 28.856 (1.961) | 29.537 (0.930) | 29.283 (1.144) | 0.361 (1.466) | | |
| WDT thenar (°C) | Benfotiamine (24) | 36.605 (4.195) | 37.043 (4.430) | 36.886 (4.047) | −0.111 (1.422) | 0.508 (-0.517 to 1.532) | 0.328 |
| | Placebo (27) | 36.626 (3.818) | 35.863 (2.913) | 35.974 (2.824) | −0.657 (2.160) | | |
| CDT foot (°C) | Benfotiamine (24) | 19.000 (10.360) | 17.770 (9.405) | 18.362 (9.824) | 1.400 (6.869) | 0.302 (-4.662 to 5.265) | 0.903 |
| | Placebo (27) | 19.781 (8.603) | 21.719 (7.983) | 19.700 (8.836) | 0.732 (8.624) | | |
| WDT foot (°C) | Benfotiamine (24) | 43.876 (4.869) | 44.126 (5.156) | 44.038 (4.262) | 0.167 (2.999) | 0.044 (-1.990 to 2.078) | 0.966 |
| | Placebo (27) | 44.115 (4.027) | 43.477 (3.832) | 43.864 (3.738) | −0.491 (4.820) | | |

.CDT, cold detection threshold; VPT, vibration perception threshold; WDT, warmth detection threshold.

In a recent pilot trial conducted over 1 year, 22 patients with type 1 or type 2 diabetes and DSPN were administered benfotiamine 600 mg/day for 3 months, followed by 300 mg/day or placebo.[20] However, due to its premature termination, clear-cut conclusions cannot be drawn from this trial, especially in view of the very small sample size at 12 months. It is also difficult to draw any firm conclusions from the two combination trials, since it remains unclear, which component could have been responsible for the effect. Nonetheless, in the study by Stracke *et al*, only peroneal MNCV, but not median MNCV and VPT were improved after 12 weeks.[14]

One long-term study reported that benfotiamine treatment (300 mg/day) over 24 months had no effects on peripheral nerve function or circulating biomarkers of inflammation in participants with type 1 diabetes.[21] It has been argued that the number of participants included with DSPN confirmed by reduced NCV was presumably very low and, hence, this trial does not provide any information on the important question as to whether treatment with benfotiamine may improve nerve damage due to DSPN or delay its progression.[22] On the other hand, it has been proposed that patients with developing rather than established DSPN should be preferably selected in RCTs of DSPN.[23] Indeed, among the 59 patients who completed the study by Fraser *et al*,[21] 56% had abnormal NCS and 16% had probable DSPN at baseline, suggesting that benfotiamine had no favorable effect on early or mild DSPN in people with type 1 diabetes. Overall, although some previous RCTs using benfotiamine pointed to a modest efficacy of the drug on neuropathic symptoms, neuropathic deficits and nerve dysfunction were not significantly improved in the short term and long term.

It should be clarified that the lack of efficacy of benfotiamine observed herein was not due to any potential sources of bias that could have resulted in a false-negative type II error. Therefore, the following aspects require consideration:

### Comprehensive set of endpoints

Since it is not clear which components of DSPN may be most susceptible to benfotiamine treatment, an overarching set of secondary endpoints was used to capture the majority of features of peripheral and autonomic nerve dysfunction and damage. Quantitative functional assessment included NCS, which represents the gold standard to assess large nerve fiber function, while skin biopsy that allows for a direct quantification of IENFD was employed as the gold standard to detect small nerve fiber loss. In addition, novel biomarkers were obtained from skin biopsies targeting microvascular, antioxidative and inflammatory processes.[24–26] A recent meta-analysis confirmed the utility of CCM as an endpoint to identify nerve regeneration following pharmacological and surgical interventions.[27] QST included six indices, while autonomic testing included multiple parameters of HRV, BRS and sudomotor function. Finally, a variety of clinical scores for neuropathic symptoms and deficits as well as



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**Table 4** Clinical scores for neuropathic symptoms and signs

| Endpoint | Treatment groups (n) | Baseline (BL) | Month 6 | Month 12 | Difference BL vs month 12 | Treatment effect (95% CI) | P value |
|---|---|---|---|---|---|---|---|
| NDS | Benfotiamine (24)<br>Placebo (27) | 6.708 (1.367)<br>5.407 (1.803) | 6.958 (2.116)<br>5.704 (2.016) | 7.000 (2.116)<br>6.208 (2.043) | 0.136 (1.726)<br>0.792 (1.560) | –0.246 (–1.300 to 0.809) | 0.647 |
| NIS-LL | Benfotiamine (24)<br>Placebo (27) | 8.125 (3.301)<br>7.481 (3.367) | 8.417 (3.106)<br>7.556 (3.796) | 9.769 (3.419)<br>8.556 (3.745) | 0.385 (2.256)<br>0.667 (2.635) | 0.358 (–1.289 to 2.006) | 0.660 |
| MNSI (EP) | Benfotiamine (24)<br>Placebo (27) | 4.979 (1.289)<br>4.204 (1.502) | 5.083 (1.606)<br>4.370 (1.504) | 5.136 (1.390)<br>4.646 (1.833) | 0.045 (1.101)<br>0.333 (1.257) | –0.140 (–0.861 to 0.581) | 0.704 |
| mTCNS total | Benfotiamine (24)<br>Placebo (27) | 13.042 (6.025)<br>12.556 (6.222) | 12.083 (5.453)<br>10.407 (6.141) | 13.591 (7.189)<br>12.292 (7.232) | 0.045 (1.101)<br>0.333 (1.257) | 0.586 (–2.096 to 3.268) | 0.668 |
| mTCNS signs | Benfotiamine (24)<br>Placebo (27) | 6.125 (3.893)<br>6.148 (3.910) | 5.042 (3.368)<br>4.667 (3.679) | 5.818 (4.807)<br>5.958 (4.563) | –0.043 (4.172)<br>–0.083 (3.425) | –0.449 (–2.404 to 1.505) | 0.651 |
| mTCNS symptoms | Benfotiamine (24)<br>Placebo (27) | 5.042 (2.710)<br>4.370 (2.950) | 5.125 (2.724)<br>4.037 (3.402) | 6.136 (3.371)<br>4.542 (3.203) | 0.864 (2.436)<br>0.042 (2.010) | 0.992 (–0.316 to 2.300) | 0.137 |
| NSS | Benfotiamine (24)<br>Placebo (27) | 7.958 (1.268)<br>7.667 (1.177) | 7.333 (2.140)<br>7.667 (1.240) | 7.273 (2.511)<br>7.958 (1.398) | –0.565 (2.483)<br>0.333 (1.494) | –0.982 (–2.145 to 0.181) | 0.098 |
| TSS feet | Benfotiamine (24)<br>Placebo (27) | 4.146 (3.255)<br>5.236 (3.444) | 3.758 (2.944)<br>4.942 (3.991) | 4.265 (3.840)<br>5.367 (3.898) | 0.230 (4.057)<br>0.142 (2.615) | –0.691 (–2.551 to 1.170) | 0.465 |
| TSS calves | Benfotiamine (24)<br>Placebo (27) | 0.680 (1.427)<br>1.393 (2.319) | 1.193 (2.001)<br>0.851 (1.937) | 1.951 (3.928)<br>1.234 (2.739) | 1.200 (3.615)<br>–0.263 (2.583) | 0.459 (–1.338 to 2.255) | 0.616 |
| NPSI | Benfotiamine (24)<br>Placebo (27) | 14.542 (14.987)<br>18.741 (16.661) | 12.667 (17.844)<br>17.192 (17.139) | 13.636 (18.290)<br>15.875 (15.057) | 1.783 (20.562)<br>–3.083 (13.204) | 0.190 (–8.895 to 9.274) | 0.967 |
| NRS 24 hours pain | Benfotiamine (24)<br>Placebo (27) | 0.958 (1.967)<br>2.074 (2.688) | 0.542 (1.382)<br>2.000 (2.527) | 1.591 (2.772)<br>2.208 (2.963) | 0.652 (3.256)<br>0.167 (1.274) | –0.197 (–1.560 to 1.166) | 0.777 |
| NRS day pain | Benfotiamine (24)<br>Placebo (27) | 0.958 (1.967)<br>2.074 (2.688) | 0.542 (1.382)<br>2.000 (2.527) | 1.591 (2.772)<br>2.208 (2.963) | 0.652 (3.256)<br>0.167 (1.274) | –0.197 (–1.560 to 1.166) | 0.777 |
| NRS night pain | Benfotiamine (24)<br>Placebo (27) | 1.250 (2.878)<br>1.852 (2.878) | 0.458 (1.719)<br>1.630 (2.720) | 1.545 (2.756)<br>2.000 (3.388) | 0.304 (3.783)<br>0.250 (3.068) | –0.411 (–2.089 to 1.268) | 0.631 |
| BPI score severity | Benfotiamine (24)<br>Placebo (27) | 1.036 (2.200)<br>1.667 (2.099) | 1.552 (2.409)<br>0.942 (1.660) | 1.284 (2.075)<br>1.104 (1.719) | 0.750 (2.174)<br>–0.583 (2.601) | 0.238 (–0.948 to 1.425) | 0.693 |
| BPI interference score | Benfotiamine (24)<br>Placebo (27) | 1.446 (2.656)<br>1.429 (2.030) | 0.964 (1.662)<br>1.027 (1.925) | 0.825 (1.446)<br>0.929 (1.567) | 0.012 (1.686)<br>–0.524 (2.649) | –0.088(–0.975 to 0.799) | 0.846 |

BPI, brief pain inventory; MNSI (EP), Michigan Neuropathy Screening Instrument (Examination Part); mTCNS, modified Toronto Clinical Neuropathy Score; NDS, Neuropathy Disability Score; NIS-LL, Neuropathy Impairment Score – Lower Limbs; NPSI, Neuropathic Pain Symptom Inventory; NRS, numerical rating scale for pain; NSS, Neuropathy Symptom Score; TSS, Total Symptom Score.



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**Table 5** Concentrations of thiamine biomarkers in blood

| Endpoint | Treatment groups (n) | Baseline (BL) | Month 6 | Month 12 | Difference BL vs month 12 | Treatment effect (95% CI) | P value |
|---|---|---|---|---|---|---|---|
| TDP (nmol/L) | Benfotiamine (24) | 182.000 (65.487) | 376.292 (62.397) | 374.636 (61.907) | 191.091 (97.360) | 151.04 (98.895 to 203.20) | <0.001 |
| | Placebo (27) | 173.148 (55.076) | 185.667 (56.892) | 211.208 (82.801) | 40.000 (96.236) | | |
| TMP (nmol/L) | Benfotiamine (24) | 3.792 (2.187) | 10.917 (5.687) | 12.091 (6.618) | 8.182 (6.780) | 6.792 (3.636 to 9.948) | <0.001 |
| | Placebo (27) | 3.630 (2.115) | 2.778 (1.649) | 4.000 (2.359) | 0.292 (3.432) | | |
| Free thiamine (nmol/L) | Benfotiamine (24) | 9.208 (21.569) | 366.042 (97.365) | 336.273 (141.996) | 326.727 (149.590) | 288.88 (202.53 to 375.22) | <0.001 |
| | Placebo (27) | 5.519 (2.666) | 6.556 (9.275) | 22.208 (80.500) | 16.875 (80.802) | | |
| Total thiamine (nmol/L) | Benfotiamine (24) | 194.917 (67.022) | 390.375 (47.153) | 378.045 (57.917) | 181.136 (91.925) | 144.58 (93.866 to 195.30) | <0.001 |
| | Placebo (27) | 182.259 (55.675) | 195.037 (56.689) | 220.458 (81.833) | 40.208 (95.689) | | |
| Transketolase activity (U/L) | Benfotiamine (24) | 62.675 (13.509) | 76.338 (17.575) | 75.610/ (14.060) | 13.815 (12.879) | 16.128 (6.086 to 26.170) | 0.003 |
| | Placebo (27) | 58.585 (9.577) | 63.130 (13.868) | 55.633 (11.701) | −3.542 (13.781) | | |
| TPP effect (%) | Benfotiamine (24) | 12.975 (7.915) | 4.217 (5.008) | 3.035 (2.546) | −10.045 (8.598) | −8.626 (−13.72 to −3.535) | 0.002 |
| | Placebo (27) | 13.667 (7.063) | 13.104 (6.739) | 13.313 (8.669) | −0.208 (10.640) | | |

TDP, thiamine diphosphate; TMP, thiamine monophosphate; TPP, thiamine pyrophosphate.

measures of well-being were applied. Since none of these overarching DSPN outcomes was favorably influenced by benfotiamine, it can be excluded that a potential benefit on other DSPN components was missed.

### Mono-center study design

To more clearly demonstrate disease progression in RCTs of DSPN, it has been proposed that, among others, a restricted number of centers and expert examiners (trained, certified, using standard approaches and reference values and interactive surveillance of tests) are used. Assuming that DSPN worsens very slowly and that adjuvant treatments are unlikely to improve nerve function to a greater degree than from euglycemia, endpoints needed to detect a difference between treatment and placebo must have excellent clinimetric characteristics.[23] The mono-center study design of the present study ensured a high-quality standard of all measurements without any intercenter bias.

### Mild-to-moderate DSPN

It has been hypothesized that with increasing duration of diabetes, the reversible (metabolic) component of DSPN gets proportionally smaller and the irreversible (structural) component gets proportionally larger. Therefore, it has been suggested that patients with mild-to-moderate neuropathy would constitute the preferred group to be included in RCTs,[28] which was taken into account herein. Although symptomatic DSPN was an inclusion criterion, the severity of neuropathic symptoms, especially pain, was relatively low. Of note, the NSS[29] at baseline was relatively high when indirectly compared with the TSS and NRS for 24-hour pain. An explanation for this apparent discrepancy could be that, unlike the TSS and NRS, the NSS classifies symptoms as being typical for DSPN[30] rather than rating their severity. It is conceivable that the trend towards improvement in the NSS observed for benfotiamine after 12 months was due to the relatively high baseline intensity leaving more room for improvement. In contrast, the baseline levels of the TSS, NRS and BPI were relatively low, thus limiting the potential for clinically relevant improvement.

### Sample size

The study was powered to detect a difference in CNFL as the primary endpoint between the groups after 12 months of treatment. However, it is conceivable that the calculated sample size was not sufficient to provide the statistical power to detect significant differences for the secondary endpoints. Since no trends in favor of benfotiamine were found for the multiple other endpoints, except for the NSS, the sample size alone unlikely explains the lack of an effect of benfotiamine.

### Study duration

It has been suggested that, apart from a placebo effect on symptoms or signs and measurement noise, the main reason why it is difficult to demonstrate monotonic worsening of neuropathic endpoints appears to be a very



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slow worsening of DSPN.[23] Given the slow progression of DSPN particularly in the RCT setting, to demonstrate drug efficacy within 1 year, regression of peripheral nerve abnormalities rather than mere slowing of progression would be anticipated. Since nerve function and structure did not deteriorate to a meaningful degree with placebo treatment in the present study, it cannot be excluded that the duration of the present trial was too short to show any improvement. If the drug is able to delay the progression of DSPN, several years would be required to demonstrate efficacy. Indeed, the Neurological Assessment of Thioctic Acid in Diabetic Neuropathy (NATHAN) 1 trial showed that more patients achieved a clinically meaningful improvement and fewer showed progression of neuropathic deficits with alpha-lipoic acid than with placebo only after 4 years but not after 2 years of treatment.[31]

### Potential confounders

Glycemic control was good and stable throughout the study without any deterioration or differences between the groups. Likewise, body weight did not differ between the groups and remained stable throughout the study without a relevant increase. The percentages of patients taking gabapentin, pregabalin or duloxetine and the rates of those who received paracetamol as rescue medication did not differ between the groups. Overall, no bias by potentially confounding variables was identified, except for a lower proportion of female participants (17.5%).

### Drug monitoring

After 12 months, all thiamine biomarker levels increased significantly from baseline to 12 months in the group administered benfotiamine as compared with the placebo group. Since four thiamine biomarker levels increased also in the placebo group (especially free thiamine), it cannot be ruled out that at least one patient on placebo had taken benfotiamine. Overall, all participants had a treatment compliance >80%. Of note, most participants in both groups had levels for the vitamin B1 analytes at baseline within the normal reference ranges. Only five participants (placebo: n=3; benfotiamine: n=2) had TDP levels slightly below the reference range. Therefore, it is possible that the apparently good supply of patients with thiamine and thus lack of thiamine deficiency at baseline may have contributed to the negative outcome of this trial.

### Dropouts

There were 10 dropouts before the 12-month visit (five in each group), 6 of whom were not included in the ITT analysis, since they did not have the primary endpoint assessed. Given the fact that the COVID-19 pandemic started during the recruitment phase of the trial and the trial duration of 1 year, the dropout rate of 17.5% was reasonable.

### COVID-19 pandemic

With the COVID-19 outbreak starting in March 2020 and the global decrease in activity of clinical trials,[32] the duration of the present trial was significantly prolonged. Furthermore, due to hygiene restrictions by the government the COVID-19 pandemic also resulted in several protocol deviations. In particular, several shorter and a few stronger shifts in visits could generally have contributed to higher data variability. In addition, the severe restrictions that participants have experienced during the pandemic could likely have exerted a negative impact on their quality of life.

The study has several limitations such as possibly a too short duration, relatively small sample size and potential unknown bias due to the COVID-19 pandemic. Among these, the study duration may be of critical importance when interpreting the results, since the progression of DSPN is rather slow or even absent, especially in the setting of an RCT. Therefore, it cannot be excluded that the duration of this trial was too short to demonstrate a clinically meaningful effect on nerve dysfunction and pathology by benfotiamine treatment in DSPN.

In conclusion, this phase II mono-center, double-blinded, placebo-controlled RCT did not demonstrate that treatment with benfotiamine administered 300 mg orally two times per day over 12 months had a favorable effect on morphometric, functional and clinical neuropathic endpoints as well as quality of life when compared with placebo treatment in people with type 2 diabetes and mild-to-moderate symptomatic DSPN. These findings do not contradict the use of benfotiamine for treatment of neuropathies caused by vitamin B1 deficiency.[33 34] In the future, larger scale RCTs conducted over several years, arguably in individuals selected for low thiamine levels as we recently suggested,[35] are needed to fully explore the potential of benfotiamine in slowing or halting the progression of DSPN.

**Author affiliations**
[1]Institute for Clinical Diabetology, German Diabetes Center (DDZ), Leibniz Center for Diabetes Research at Heinreich Heine University, Düsseldorf, Germany
[2]German Center for Diabetes Research (DZD e.V.), Partner Düsseldorf, Munich-Neuherberg, Germany
[3]Institute for Biometrics and Epidemiology, German Diabetes Center (DDZ), Leibniz Center for Diabetes Research at Heinrich Heine University, Düsseldorf, Germany
[4]Department of Endocrinology and Diabetology, Medical Faculty and University Hospital, Heinrich Heine University, Düsseldorf, Germany
[5]Institute for Clinical Biochemistry and Pathobiochemistry, German Diabetes Center (DDZ), Leibniz Institute for Diabetes Research at Heinrich Heine University, Düsseldorf, Germany
[6]Wörwag Pharma GmbH & Co. KG, Böblingen, Germany
[7]Institute for Health Services Research and Health Economics, German Diabetes Center (DDZ), Leibniz Center for Diabetes Research at Heinrich Heine University, Düsseldorf, Germany
[8]Institute for Health Services Research and Health Economics, Centre for Health and Society, Medical Faculty and University Hospital, Heinrich Heine University, Düsseldorf, Germany
[9]Centre for Health and Society, Medical Faculty and University Hospital, Heinrich Heine University, Düsseldorf, Germany

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**Contributors** DZ, GJB and CR conceived and refined the trial design. DZ, AS, CH, BK, AI, HA-H, CR and MR compiled the protocol for assessments. GJB, GS and DZ conducted the trial. KS and OK developed the statistical analysis plan. OK and LJ performed the statistical analysis. DZ drafted the manuscript. All authors critically contributed to the writing and approved the final manuscript for publication. DZ is the guarantor of this work.

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**Competing interests** DZ and GJB received honoraria from Wörwag Pharma for speaking and consulting activities. CR, AM and BAJ are employees of Wörwag Pharma. MR received research support from Boehringer-Ingelheim and Novo Nordisk and honoraria for speaking and/or consulting activities from AstraZeneca, Boehringer-Ingelheim, Echosens, Lilly, Madrigal, MSD, Novo Nordisk, Sanofi, Synlab and Target RWE. GS, CH, AS, BK, AI, HA, KS, OK and LJ have no conflicts of interest to report.

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**Data availability statement** No data are available. The full trial protocol and statistical analysis plan are not publicly available, but the study design has been published[18] and can also be accessed via EudraCT (2017-003054-16) and DRKS (DRKS00014832). Anonymised patient data and statistical code are not publicly accessible.

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**ORCID iDs**

Dan Ziegler https://orcid.org/0000-0001-8956-3552
Gundega Sipola https://orcid.org/0009-0006-7638-0638
Klaus Strassburger https://orcid.org/0000-0002-6636-1373
Christian Herder https://orcid.org/0000-0002-2050-093X
Birgit Knebel https://orcid.org/0000-0002-5575-4315
Claudia Reule https://orcid.org/0009-0000-3234-6987
Andrea Carolina Moreira Tupac Yupanqui https://orcid.org/0009-0005-8153-7428
Andrea Icks https://orcid.org/0000-0002-4882-969X
Hadi Al-Hasani https://orcid.org/0000-0003-2543-0130
Michael Roden https://orcid.org/0000-0001-8200-6382
Oliver Kuss https://orcid.org/0000-0003-3301-5869
Gidon J Bönhof https://orcid.org/0000-0003-1446-6592

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